

# TP2/RR2 for SPY and SPX options

May 13, 2026

## 1 Summary statistics

The options data on SPY and SPX covers the period 1-Jan-25 to 29-Aug-25, for a total of 165 trading days.

Table 1: Violation rates for SPY and SPX options. First two rows include both AM- and PM-settled SP500 options. The rows 3,4 show the results keeping only PM-settled options (SPXW).

|            | $n_{tot}$   | $[n_{vio}]_{mid}$ | $vr_{mid}$ | $[n_{vio}]_{bid-ask}$ | $vr_{bid-ask}$ |
|------------|-------------|-------------------|------------|-----------------------|----------------|
| SPX calls  | 167,812,336 | 1,154,833         | 0.688%     | 140,591               | 0.084%         |
| SPX puts   | 429,757,716 | 10,955,227        | 2.550%     | 1,839,288             | 0.428%         |
| SPXW calls | 63,081,514  | 177,028           | 0.281%     | 3,899                 | 0.006%         |
| SPXW puts  | 144,042,772 | 3,329,029         | 2.311%     | 407,911               | 0.283%         |
| SPY calls  | 43,376,334  | 176,859           | 0.408%     | 10,909                | 0.025%         |
| SPY puts   | 62,305,648  | 1,182,236         | 1.897%     | 232,228               | 0.373%         |

The violation rates for both SPX and SPY are shown in Table 1. The  $TP_2/RR_2$  violations are computed using mid-prices. The result is consistent with Mike's Figure 22 which shows larger violations for SPX puts than calls. The difference between the violation rates for calls and puts becomes larger when restricting only to PM-settled options SPXW.

**Restricting to OTM options.** Table 2 compares the violation ratios for all strikes with those obtained by restricting only to OTM strikes. For simplicity I used the spot price  $S_0$  on the pricing date to distinguish OTM from ITM options. These results include both AM- and PM-settled options.

Theoretical expectation in the BS model with positive drift  $r - q \geq 0$  is that  $TP_2$  holds for European calls with all strikes, while  $RR_2$  holds only for European OTM puts. No such restrictions are expected for American options.

Thus, for SPX we expect to see comparable TP2 violation rates for OTM and for all- $K$ , and higher RR2 violations for all- $K$  compared to OTM puts alone. The numbers show higher TP2 violation rates for OTM calls, and slightly larger RR2 violation rates for puts (opposite to expectations).

For SPY we expect to see similar violation rates for OTM/all- $K$ . In fact we observe higher violation rates for OTM calls

Table 2: Violation rates for SPY and SPX options, comparing with their restriction to OTM strikes. Midprices are used for all cases. SPX includes both AM- and PM-settled options.

|                  | $vr_{all-K}$ | $vr_{OTM}$ |
|------------------|--------------|------------|
| SPX + SPXW calls | 0.688%       | 1.068%     |
| SPX + SPXW puts  | 2.550%       | 2.606%     |
| SPY calls        | 0.408%       | 1.080%     |
| SPY puts         | 1.897%       | 1.960%     |

## 2 Correlation SPY-SPX violation rates

Figure 1 shows the daily violation rates, comparing SPY and SPX, separately for calls and puts. The horizontal lines show the average violation rates.

Days of high violation rates for SPX coincide mostly with those of high violation rates for SPY. The correlations are similar to those obtained by Mike.

Table 3: Correlations among daily violation rates for SPY and SPX. The first two rows include all SPX options, and the last two rows keep only PM-settled options SPXW.

|                    | Pearson | Spearman |
|--------------------|---------|----------|
| Calls SPY-SPX(all) | 71.1%   | 53.6%    |
| Puts SPY-SPX(all)  | 88.5%   | 85.9%    |
| Calls SPY - SPXW   | 69.4%   | 86.2%    |
| Puts SPY - SPXW    | 86.2%   | 69.4%    |

The same data is shown as scatter plots in Figure 2 which shows better the correlation. The lines show a linear fit to the daily violation rates.

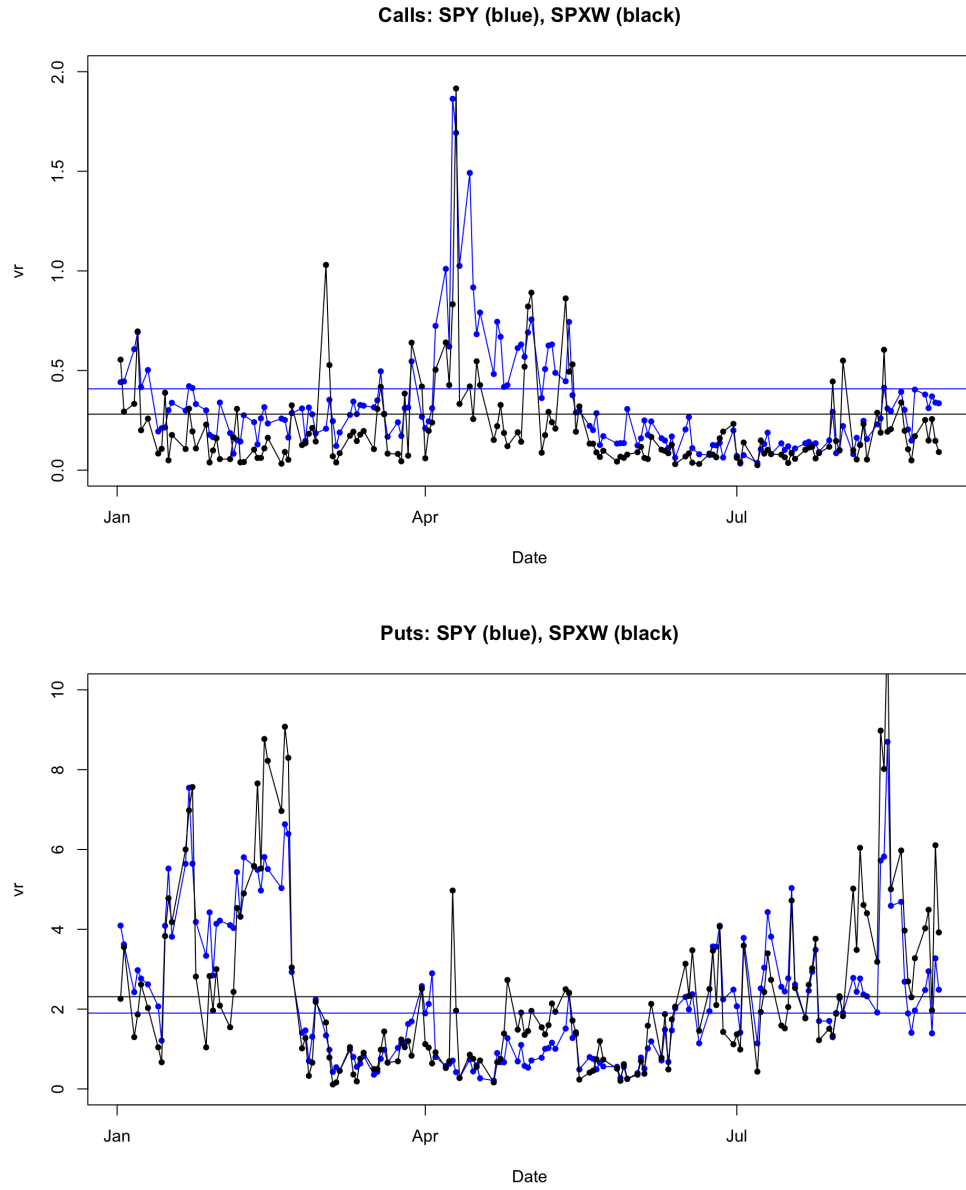


Figure 1: The violation rates for TP2/RR2 violations for each day in the 2025 dataset, comparing SPY and SPXW (PM-settle only). The horizontal lines show the average violation rates from Table 1.

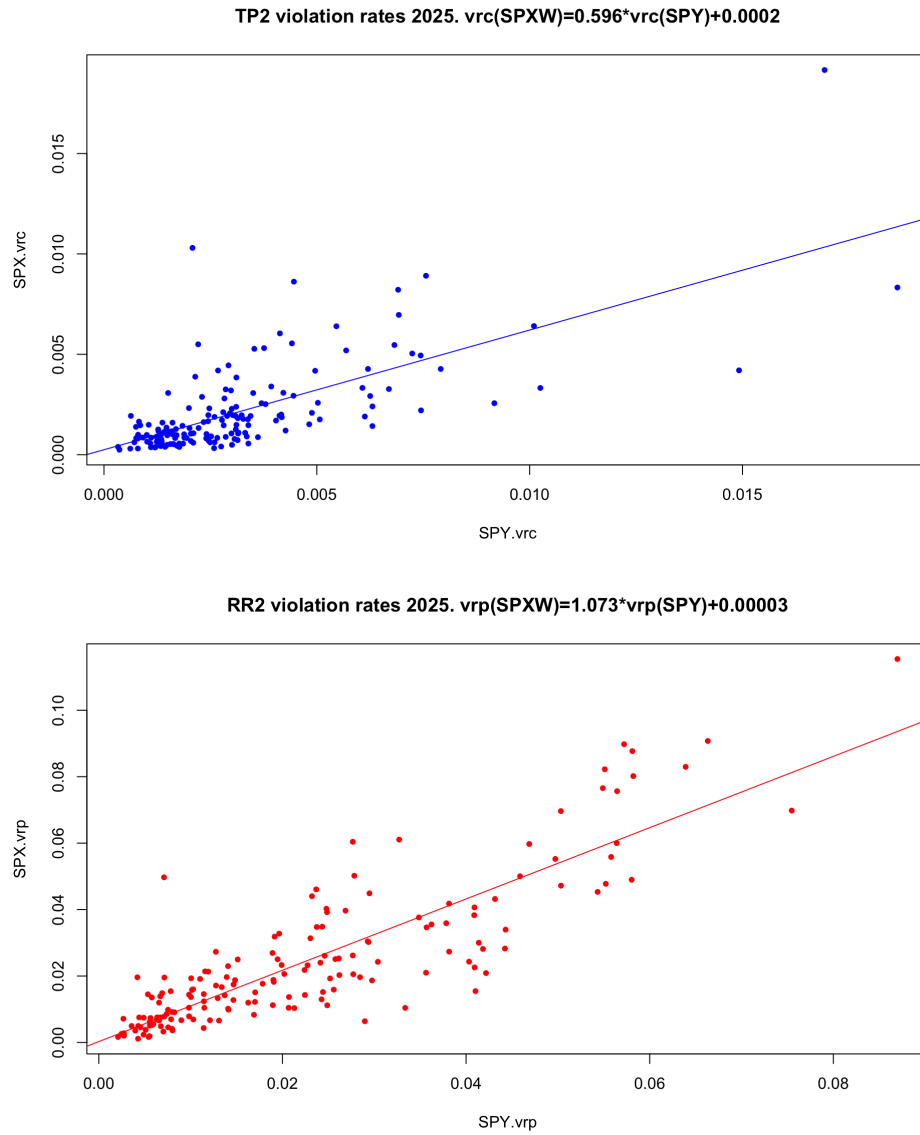


Figure 2: Scatter plots for the daily violation rates for SPXW and SPY, which show the correlation among them.